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Data Science FINAL PROJECT REPORT

Project Title:

Predictive Maintenance and Anomaly Detection in Aircraft Engines Using Multivariate Sensor Data

By

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Date: 19th April 2026

GitHub Link:
https://github.com/sairaghavendra1424/Data_Science_Project.git

Supervisor: Dr Vid Irsic

Word Count: 4767

Declaration Statement

This report is submitted in partial fulfilment of the requirements for the degree of Master of Science in Data Science at the University of Hertfordshire.

I have read and understood the University's guidelines on academic integrity, misconduct, and plagiarism, including the procedures for handling suspected cases of academic misconduct and the associated penalties, which may include failure of the project or the course.

I confirm that this report is my own work. Any material that has been derived from or quoted from published or unpublished work of others has been appropriately acknowledged. (Ref. UPR AS/C/6.1, Section 7 and UPR AS/C/5, Section 3.6)

I confirm that no human participants were involved in this MSc project.

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Acknowledgement

This project was completed as part of my course, and it shows the learning, time, and effort I have given during my studies. I worked on this project step by step, and it helped me understand the subject in a better and more practical way.

I thank my supervisor, Vid Irsic, for guiding me during this project and helping me when I had doubts. His suggestions helped me improve my work and move in the right direction. I also thank the University of Hertfordshire for giving me the chance to learn and for providing the resources needed to complete this project.

I also want to recognise my own hard work, time, and focus that I put into this project. I stayed committed to the work and continued learning throughout the process, which helped me complete this project successfully.

Abstract

Predictive maintenance has become an important approach in modern industrial systems, especially in aerospace applications where reliability and safety are critical. This project focuses on predicting the Remaining Useful Life (RUL) of aircraft engines and identifying anomaly patterns using multivariate sensor data from the NASA CMAPSS dataset. The study aims to compare statistical and machine learning methods for modelling engine degradation and supporting maintenance decisions.

A structured data science pipeline was implemented, including data preprocessing, feature selection, normalization, and feature engineering. Multiple regression models were developed for RUL prediction, including Linear Regression, Random Forest, Gradient Boosting, and XGBoost. In addition, anomaly detection techniques such as Principal Component Analysis (PCA) and Isolation Forest were applied to identify abnormal behaviour in sensor data.

The results show that Linear Regression achieved the best performance in terms of RMSE (32.04), while Gradient Boosting achieved the lowest MAE (23.22). More complex models such as Random Forest and XGBoost did not significantly outperform simpler models, indicating that the dataset contains relatively structured and partially linear relationships. Feature engineering techniques did not improve performance and, in some cases, increased prediction error. Hyperparameter tuning provided only minor improvements.

Anomaly detection results indicate that abnormal patterns increase as engine degradation progresses, suggesting that both PCA and Isolation Forest are effective in identifying potential failure behaviour. However, due to the absence of labelled anomalies, the evaluation remains exploratory.

Overall, this study demonstrates that effective predictive maintenance can be achieved using both statistical and machine learning approaches. It highlights the importance of understanding data characteristics and shows that simpler models can perform competitively in structured datasets.

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1 Introduction:

Aircraft engines are highly complex mechanical systems that operate under extreme environmental and operational conditions. Failures in such systems can lead to significant financial losses, safety risks, and operational disruptions. Therefore, ensuring the reliability and timely maintenance of aircraft engines is a critical concern in the aviation industry. Traditionally, maintenance strategies have been based on fixed schedules or reactive approaches, where repairs are carried out after a failure occurs. However, these methods are often inefficient, as they either lead to unnecessary maintenance or fail to prevent unexpected breakdowns. As shown in Figure 1, multiple sensors are placed across engine components to monitor operational parameters in real time

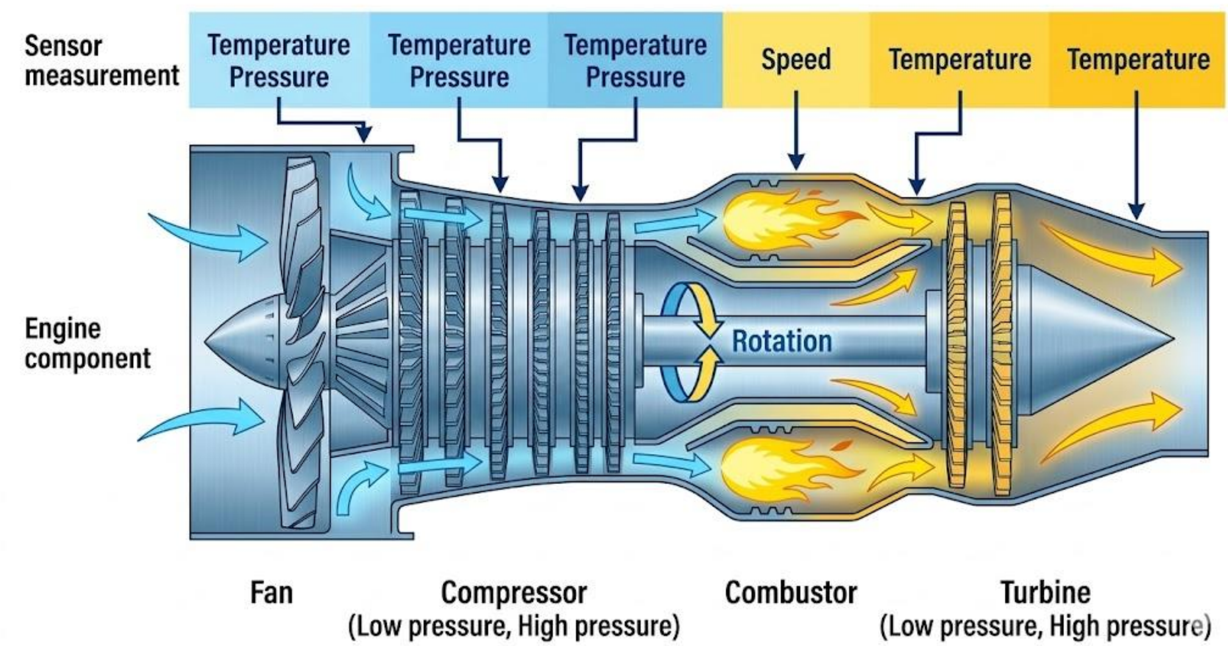

Figure 1: Aircraft Engine Sensor Measurement and Component Structure (adapted from MDPI Sensors, 2020)

In recent years, predictive maintenance has emerged as a more effective approach by using data-driven techniques to monitor system health and predict potential failures before they occur. This approach relies on continuous monitoring of sensor data collected from aircraft engines during operation. By analysing this data, it is possible to detect early signs of degradation and take preventive actions. The availability of large-scale sensor data has enabled the use of machine learning methods, which can identify complex patterns and relationships that are difficult to capture using traditional statistical techniques (Saxena et al., 2008). Figure 2 illustrates the main components of a turbofan engine, highlighting the stages where degradation and faults may occur.

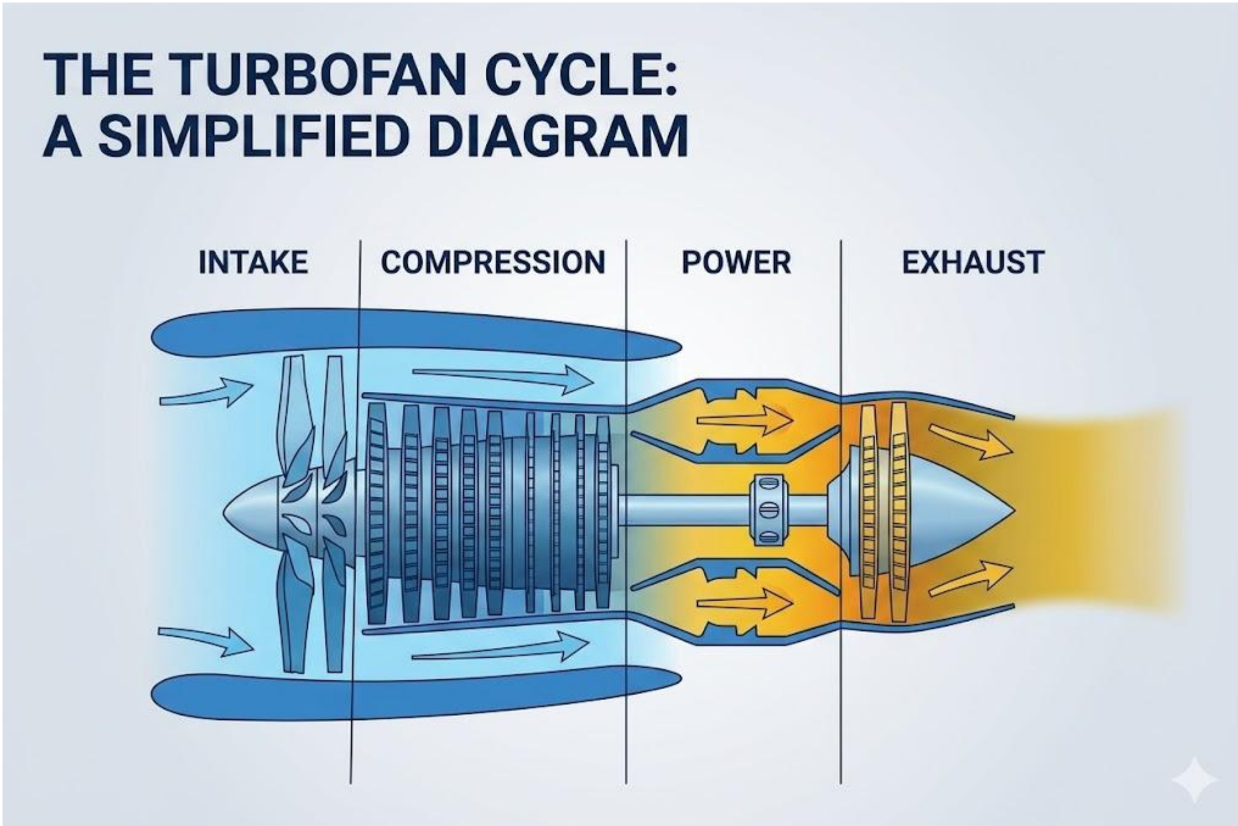

Figure 2: Main Components of a Turbofan Engine (adapted from Aloha Grace Blog, 2019)

This project focuses on predictive maintenance and anomaly detection in aircraft engines using multivariate sensor data. The dataset used in this study is provided by the NASA Prognostics Center of Excellence and consists of simulated turbofan engine run-to-failure data. It contains multiple sensor measurements recorded over time under different operating conditions, making it suitable for time-series analysis and health monitoring applications . The multivariate nature of the dataset allows for a detailed analysis of engine behaviour and degradation patterns.

Anomaly detection plays a key role in predictive maintenance, as it involves identifying unusual patterns in data that may indicate faults or abnormal system behaviour. These anomalies often represent early warning signs of potential failures. Both statistical methods and machine learning models can be used for anomaly detection. Statistical techniques are generally simple and interpretable, while machine learning methods are more powerful in handling high-dimensional and complex data (Chandola, Banerjee and Kumar, 2009).

Recent research has shown that machine learning approaches, including ensemble methods and deep learning models, can significantly improve the accuracy of anomaly detection in industrial systems. These methods are capable of learning from historical data and detecting subtle deviations in sensor patterns that may not be visible through traditional methods (Zhao et al., 2019). However, the effectiveness of these models depends on the nature of the data and the type of anomalies present.

The aim of this project is to compare statistical and machine learning-based anomaly detection methods in identifying degradation patterns in aircraft engine sensor data. By evaluating different approaches, this study seeks to determine which methods provide better performance in detecting anomalies and supporting predictive maintenance. The findings of this project can contribute to improving maintenance strategies, reducing operational costs, and enhancing the safety and reliability of aircraft systems.*The key factors influencing house prices are illustrated in Figure 1.*

2 Literature Review:

Predictive maintenance has become an important research area in industrial and aerospace systems due to its ability to reduce failures and improve system reliability. Unlike traditional maintenance strategies, predictive maintenance uses real-time sensor data to monitor system health and identify potential failures before they occur. In aircraft engines, this approach is especially important because of high operational costs, strict safety requirements, and the serious consequences of unexpected failures. Modern aircraft engines are equipped with multiple sensors that continuously record parameters such as temperature, pressure, and vibration. These multivariate sensor measurements provide valuable time-series data that can be analysed to understand system degradation and support maintenance decisions. Figure 8 illustrates the differences between preventive and condition-based maintenance, showing how predictive approaches reduce maintenance costs and improve efficiency.

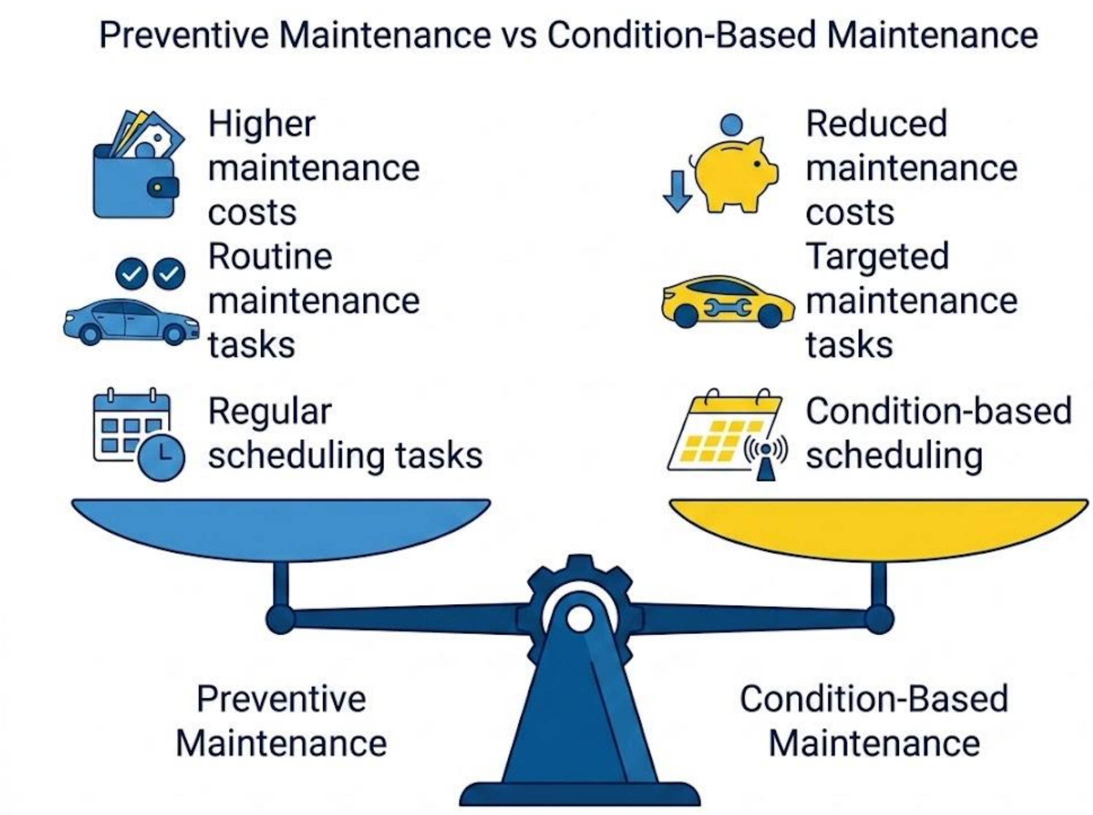

Figure 3: Preventive vs Condition-Based Maintenance (adapted from WorkTrek, 2023).

One of the key concepts in predictive maintenance is Remaining Useful Life (RUL) estimation. RUL refers to the number of operational cycles remaining before a system fails. It is commonly treated as a regression problem where the goal is to predict how long an engine can continue operating under current conditions. Accurate RUL estimation is essential for maintenance scheduling, resource planning, and risk reduction. Earlier approaches to RUL estimation were based on statistical degradation models and reliability theory. However, due to the complex and nonlinear nature of sensor data, machine learning methods have become more effective in recent studies.

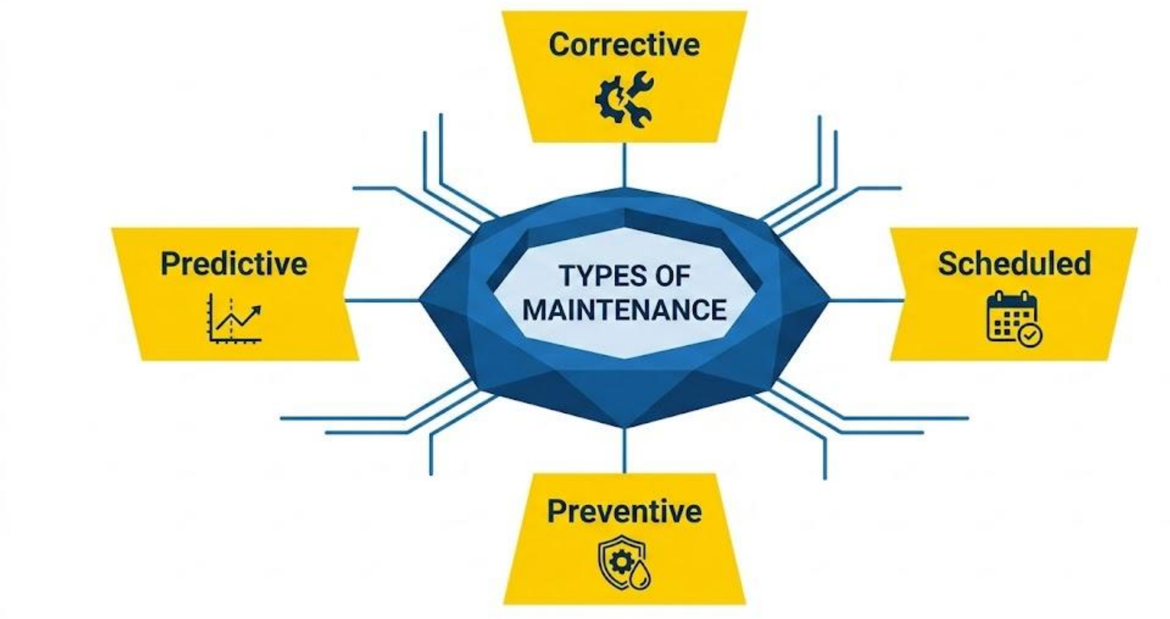

Figure 4: Types of Maintenance Strategies (adapted from Preventive Maintenance Procedure, 2025).

The NASA Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset is widely used as a benchmark for research in predictive maintenance and prognostics. This dataset was developed to simulate turbofan engine degradation under controlled conditions. Each engine starts in a healthy state and gradually degrades until failure. The dataset contains multivariate time-series data, including multiple sensor readings and operational settings. The FD001 subset, which is commonly used for baseline studies, includes 100 engines operating under a single condition and fault mode, making it suitable for controlled experimentation. Preprocessing steps such as assigning labels, generating RUL values, removing irrelevant features, and normalising data are standard practices to ensure consistency and reliability in modelling.

Statistical methods have traditionally been used for anomaly detection in industrial systems. These methods assume that normal system behaviour follows a stable distribution, and any deviation from this pattern indicates an anomaly. Techniques such as Z-score analysis, moving averages, and control charts are commonly used. Principal Component Analysis (PCA) is particularly useful in multivariate systems, as it reduces dimensionality while preserving important variance in the data. Although statistical methods are simple, interpretable, and computationally efficient, they often struggle to capture nonlinear relationships and complex dependencies present in high-dimensional sensor data (Chandola, Banerjee and Kumar, 2009).

To overcome these limitations, machine learning methods have been widely adopted in predictive maintenance. Classical models such as linear regression, random forest, support vector machines, and gradient boosting are commonly used to model relationships between sensor data and system degradation. These models are capable of handling nonlinear patterns and interactions between variables. In addition, deep learning approaches have further improved performance by capturing temporal dependencies in sequential data. Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, are particularly effective for time-series data as they can learn long-term dependencies (Zhao et al., 2019).

Autoencoder-based models are also widely used for anomaly detection. These models learn a compressed representation of normal system behaviour and identify anomalies based on reconstruction error. When the model fails to accurately reconstruct input data, it indicates that the pattern is different from normal behaviour. While machine learning and deep learning methods provide higher accuracy, they require careful preprocessing, parameter tuning, and higher computational resources compared to statistical methods.

Despite the progress in this field, there is still a gap in research regarding direct comparisons between statistical and machine learning approaches under consistent experimental conditions. Many studies focus mainly on improving prediction accuracy without considering factors such as interpretability and computational efficiency, which are important in real-world aerospace applications. This project aims to address this gap by comparing different approaches for anomaly detection and degradation analysis using the same dataset and preprocessing pipeline. By combining both statistical and machine learning methods, this study provides a more comprehensive understanding of predictive maintenance techniques for aircraft engines.

3 Methodology:

This section explains the complete approach followed in this project for predictive maintenance and anomaly detection using multivariate sensor data from aircraft engines. The methodology is based on a structured data science pipeline, including data loading, preprocessing, feature engineering, and model development.

3.1 System Overview

The project follows a sequential workflow starting from data acquisition to model building. The NASA CMAPSS dataset is first loaded and structured into a usable format. Preprocessing steps are applied to clean and transform the data, followed by feature engineering to extract meaningful patterns. Machine learning models are then developed to predict Remaining Useful Life (RUL) and identify degradation behaviour. Additionally, anomaly detection techniques are applied to detect abnormal patterns in engine performance.

This structured pipeline ensures that all models are trained on consistent data and that the analysis reflects real-world predictive maintenance scenarios.

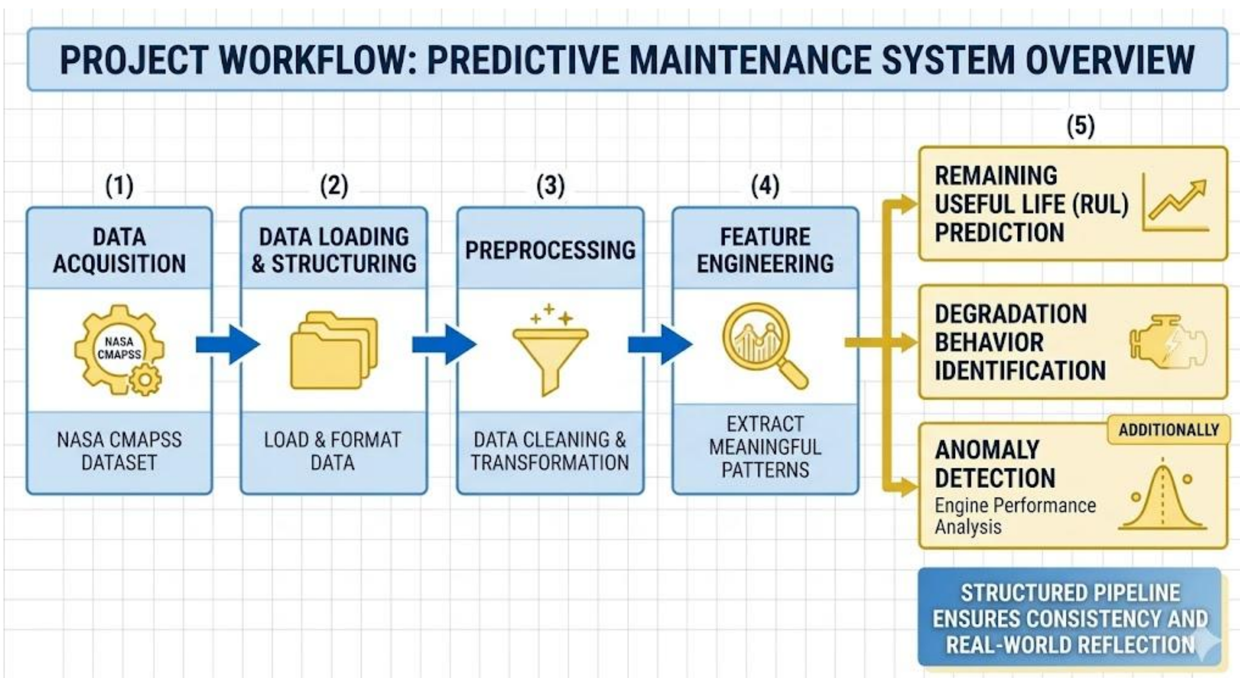

Figure 5: Predictive maintenance system overview (author's own)

3.2 Data Loading and Structure

The NASA CMAPSS FD001 dataset is used in this project. The dataset consists of three main components: training data containing full run-to-failure engine cycles, test data with partial engine sequences, and corresponding RUL values for the test dataset.

Each record in the dataset represents a single operational cycle of an engine. The dataset includes engine identifiers, cycle count, three operational settings, and twenty-one sensor measurements. Column labels are assigned to clearly define these variables, improving readability and usability for further analysis.

The training dataset is used to learn degradation patterns, while the test dataset is used to evaluate model performance on unseen engines. This separation ensures a realistic predictive maintenance setup.

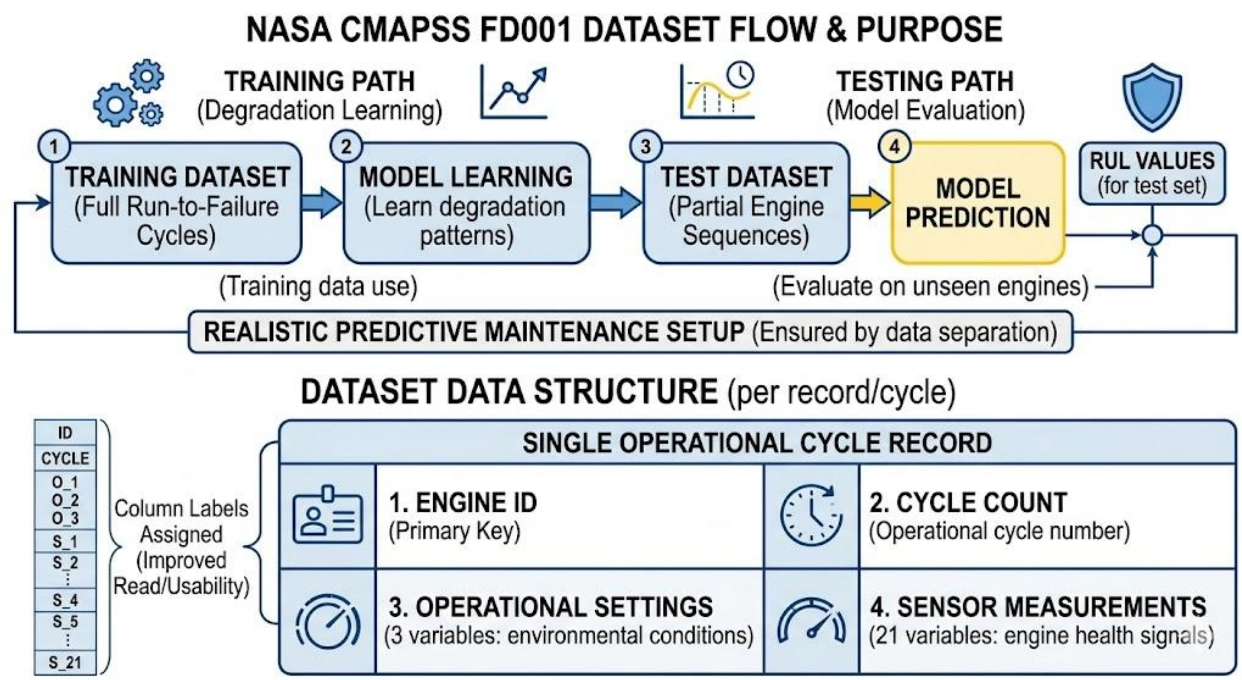

Figure 6: Data Loading and Structure

3.3 Remaining Useful Life (RUL) Calculation

The dataset does not directly provide RUL values for training data, so they are calculated manually. For each engine, the maximum number of cycles is identified, and the RUL is computed as the difference between this maximum cycle and the current cycle.

This transformation converts the dataset into a supervised regression problem, where the goal is to predict how many cycles remain before engine failure. This step is critical because it defines the target variable used for model training.

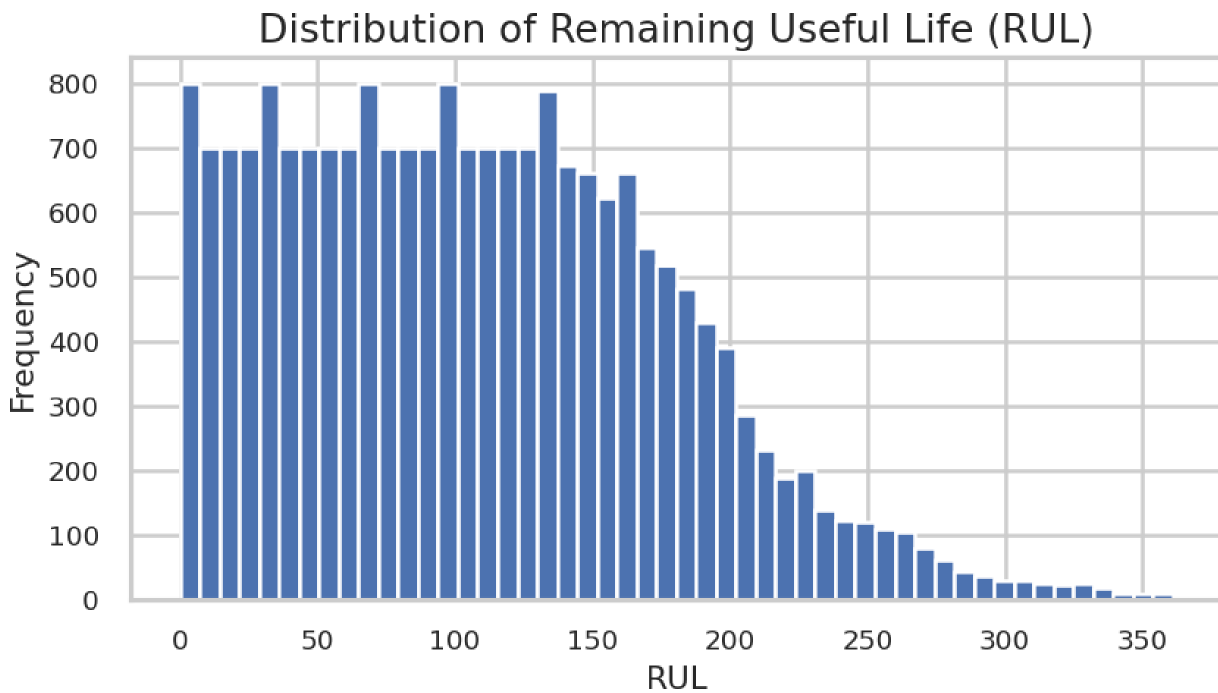

Figure 7: Distribution of Remaining Useful Life (RUL)

3.4 Data Preprocessing

Data preprocessing is performed to improve data quality and remove irrelevant information. The dataset is first checked for missing values, and it is observed that no missing values are present.

Next, the uniqueness of each feature is analysed to identify non-informative or constant features. Several sensor and operational variables that do not vary significantly are removed, including selected sensor measurements and one operational setting. Removing these features reduces noise and improves model efficiency.

The dataset is then prepared for modelling by selecting relevant feature columns and separating input features from the target variable. This step ensures that only meaningful variables are used for learning.

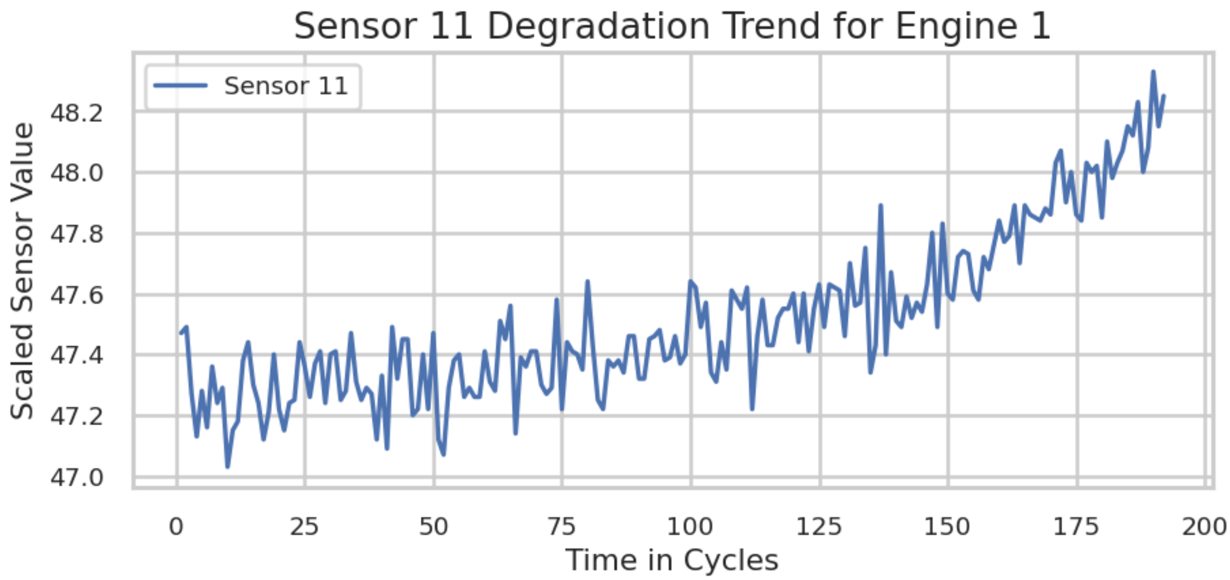

Figure 8: Sensor 11 Degradation for Engine 1

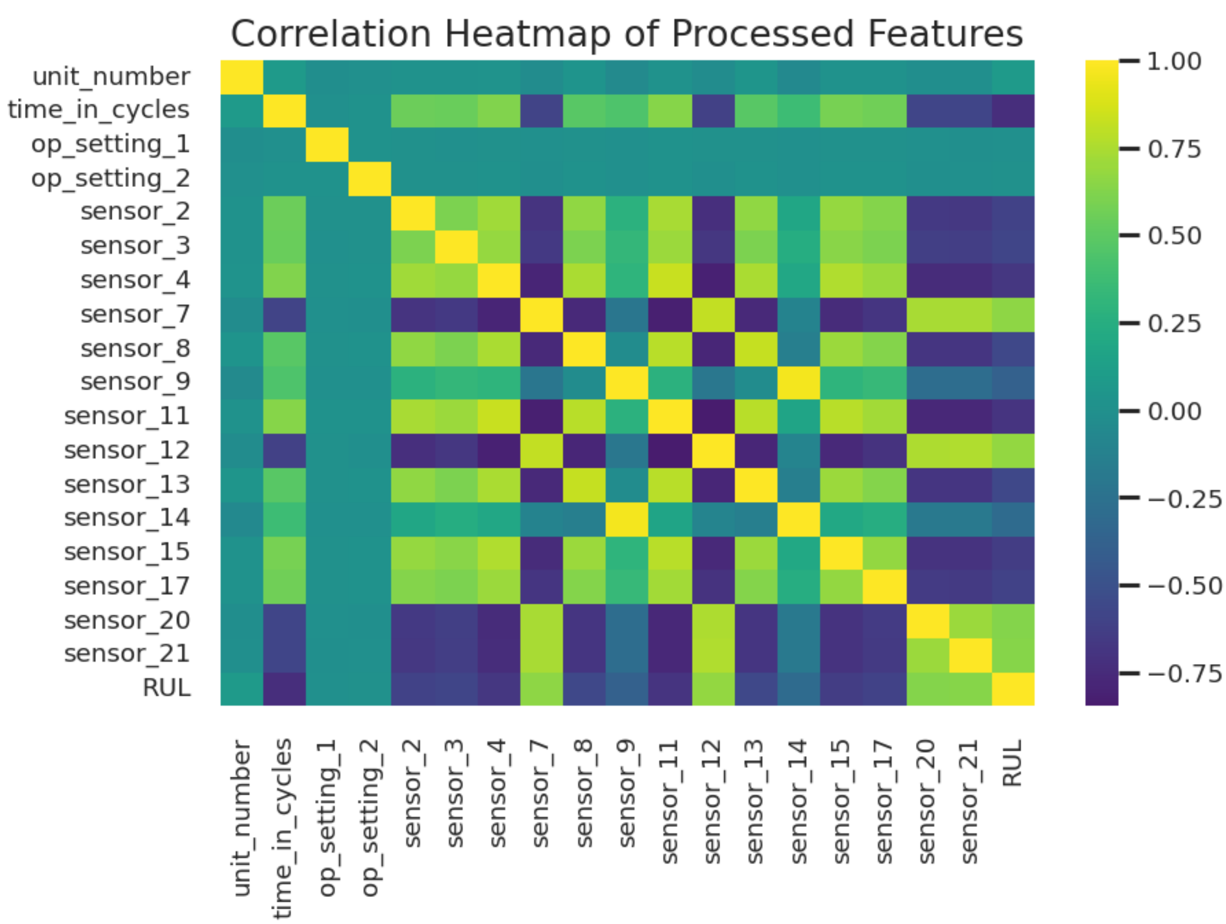

Figure 9: Correlation Heatmap of processed Features

3.5 Feature Scaling

Since sensor measurements have different value ranges, feature scaling is applied to normalize the data. MinMax scaling is used to transform all feature values into a range between 0 and 1.

This normalization ensures that no single feature dominates the model due to its scale. It also improves model stability and helps machine learning algorithms converge more effectively during training.

3.6 Exploratory Data Analysis

Exploratory Data Analysis is conducted to understand patterns in the dataset. Sensor trends are analysed over engine cycles to observe degradation behaviour. It is observed that certain sensor values change consistently as the engine approaches failure.

The distribution of RUL is also analysed to understand how remaining life varies across the dataset. Correlation analysis is performed to identify which features are strongly related to RUL. These insights confirm that the dataset contains meaningful patterns that can be used for predictive modelling.

Additionally, Principal Component Analysis is applied to visualize the data in reduced dimensions. This helps in understanding the overall structure and variation in sensor data.

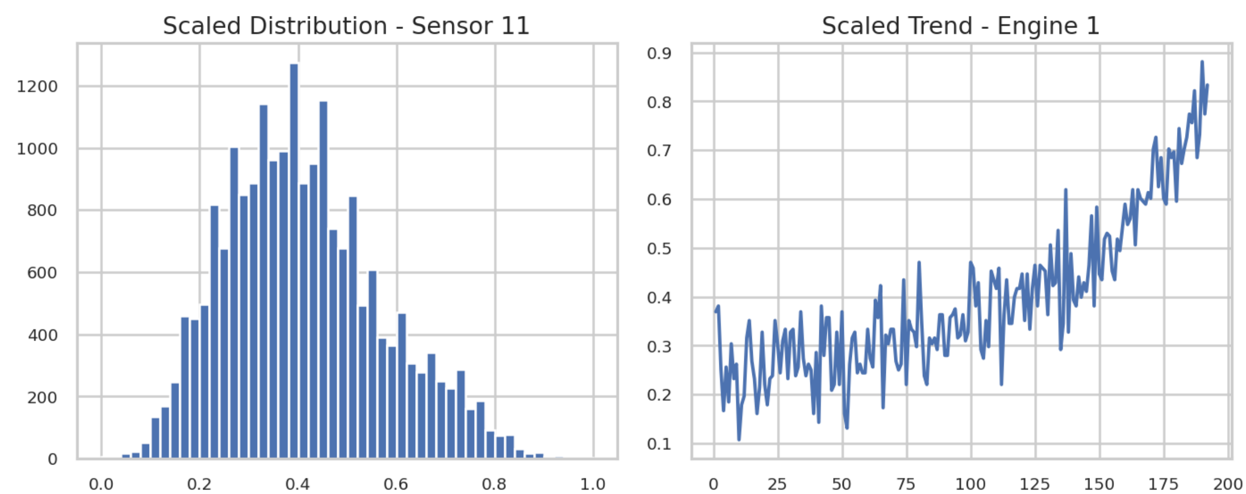

Figure 10: Scaled distribution comparision

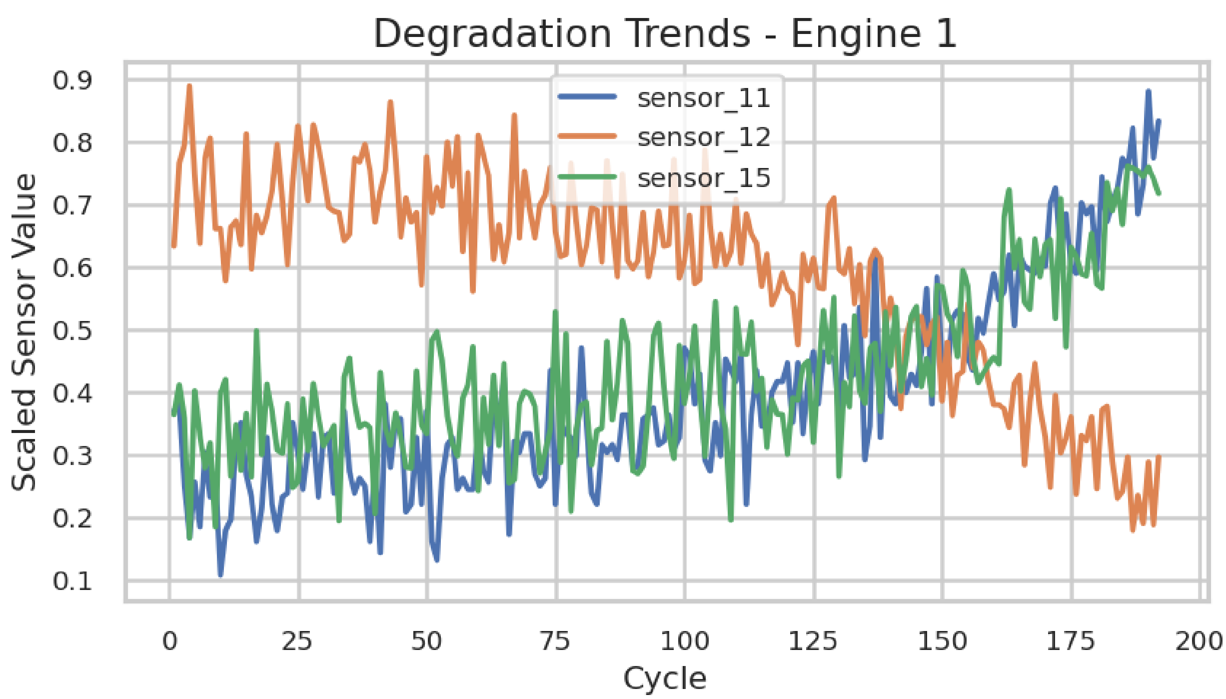

Figure 11: Degradation Trends- Engine 1

3.7 Train and Test Preparation

Instead of using a random split, the official CMAPSS train-test split is used. The training dataset contains full engine lifecycles, while the test dataset contains only the last observed cycle of each engine.

For the test dataset, the final cycle of each engine is extracted, and the corresponding RUL values are assigned using the provided ground truth. This approach prevents data leakage and ensures that models are evaluated on completely unseen engine data.

3.8 Model Development

Multiple regression models are developed to predict RUL. These include Linear Regression, Random Forest, Gradient Boosting, and XGBoost. Linear Regression is used as a baseline model, while ensemble methods such as Random Forest and boosting models are used to capture nonlinear relationships in the data.

The models are trained using the training dataset, where input features represent sensor readings and the target variable represents RUL. Each model learns the relationship between engine condition and its remaining life.

3.9 Feature Engineering

To improve model performance, additional features are created to capture temporal patterns in the data. Lag features are generated by using sensor values from previous cycles. Rolling mean features are also computed to capture short-term trends in sensor behaviour.

These engineered features help incorporate time-dependent information into the model, even though the models themselves are not sequential in nature. Missing values created due to shifting operations are handled using backward filling and default values.

After feature engineering, scaling is again applied to ensure consistency across all features.

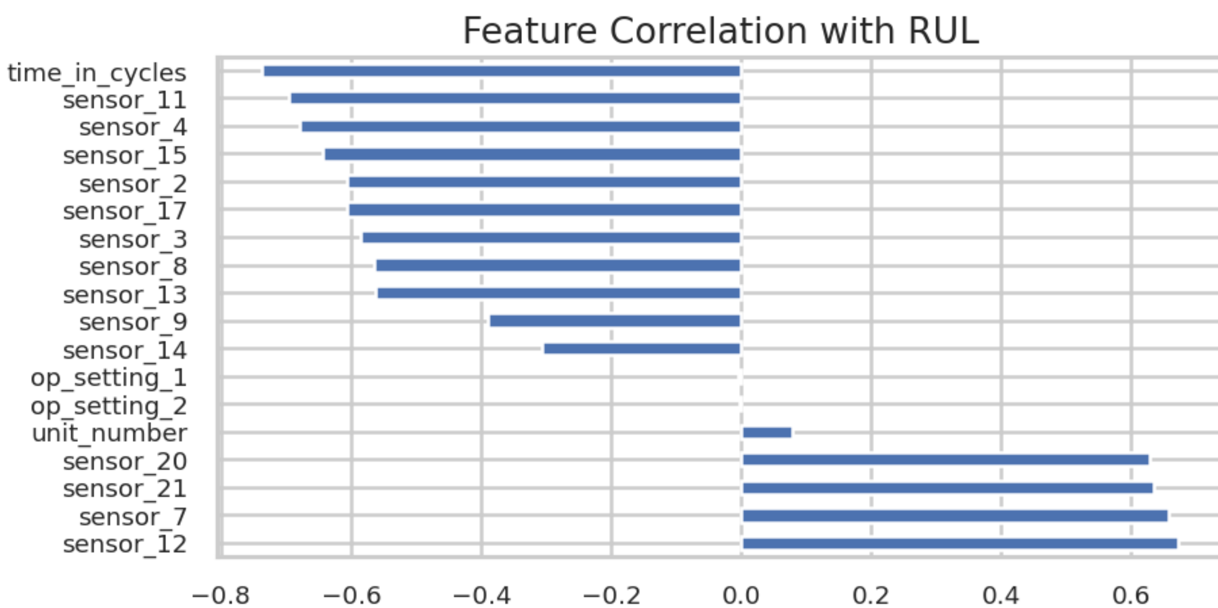

Figure 12: Feature Correlation with RUL

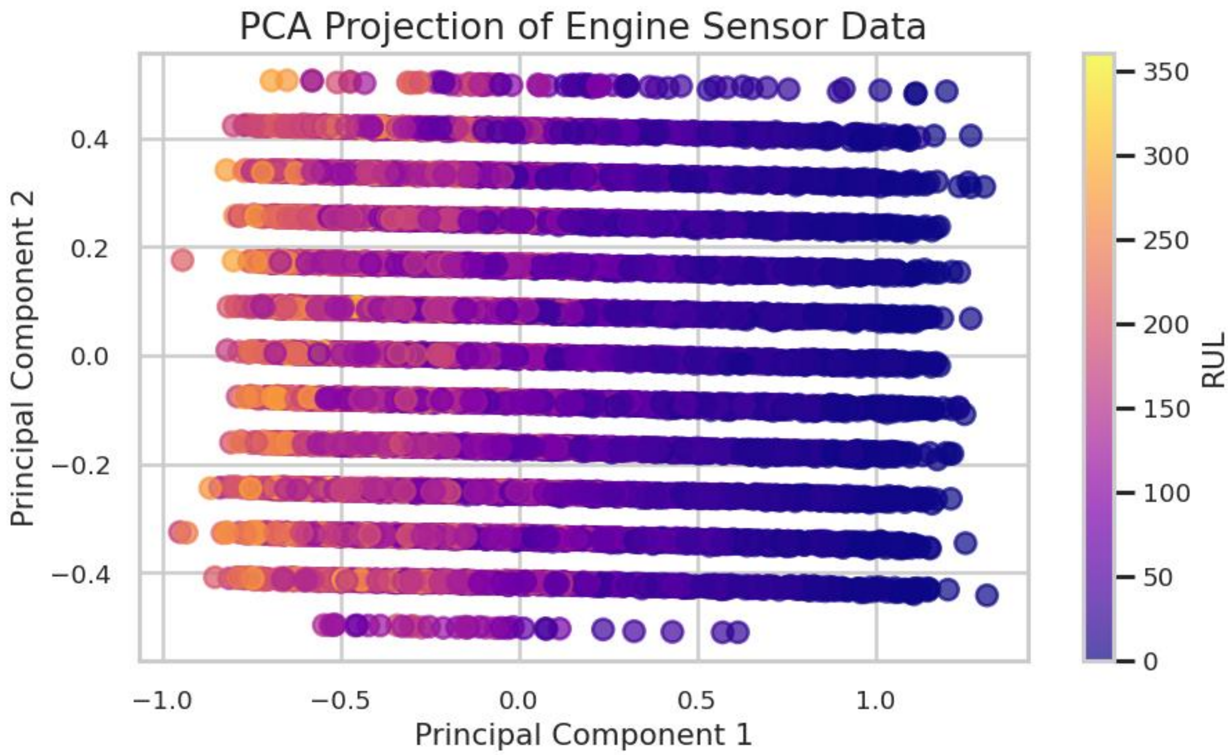

Figure 13: PCA Projection of Engine Sensor Data

3.10 Model Enhancement with Engineered Features

The models are retrained using the engineered feature set to evaluate the impact of temporal information. Random Forest and XGBoost models are specifically used for this step, as they are capable of handling complex feature interactions.

This step helps determine whether additional temporal features improve the predictive capability of the models.

3.11 Hyperparameter Tuning

To further improve model performance, hyperparameter tuning is performed on the XGBoost model using Randomized Search. Different combinations of parameters such as number of trees, depth, learning rate, and sampling ratios are tested.

Cross-validation is used during tuning to ensure that the selected parameters generalize well across different data subsets. This step helps optimize the model without overfitting.

3.12 Anomaly Detection

Anomaly detection is performed using both statistical and machine learning approaches.

Principal Component Analysis is used as a statistical method to detect anomalies. The data is projected into a lower-dimensional space and then reconstructed. The reconstruction error is calculated for each data point, and points with high error are considered anomalies.

Isolation Forest is used as a machine learning method for anomaly detection. This algorithm isolates anomalies by randomly partitioning the data. Data points that require fewer splits to isolate are considered anomalous.

Both methods are applied to identify abnormal behaviour in engine sensor data. The anomaly scores are analysed across engine cycles to observe how abnormal patterns increase as the engine approaches failure.

4 Results

This section presents the performance of different models for predicting Remaining Useful Life (RUL) and analysing anomaly patterns using the NASA CMAPSS dataset .

4.1 RUL Prediction Model Performance

Four regression models were implemented and evaluated: Linear Regression, Random Forest, Gradient Boosting, and XGBoost. The performance of these models is presented in Table 1.

Table 1: RUL Prediction Performance (Baseline Models)

Model	**RMSE**	**MAE**
Linear Regression	32.0427	25.5938
Gradient Boosting	32.3166	23.2212
Random Forest	33.2132	24.4470
XGBoost	33.5161	24.0273

The results show that **Linear Regression achieved the lowest RMSE (32.04)**, indicating the best overall prediction accuracy in terms of squared error. However, **Gradient Boosting achieved the lowest MAE (23.22)**, suggesting better performance in minimizing absolute prediction errors.

Random Forest and XGBoost showed slightly higher RMSE values, indicating that their predictions were less accurate compared to simpler models for this dataset.

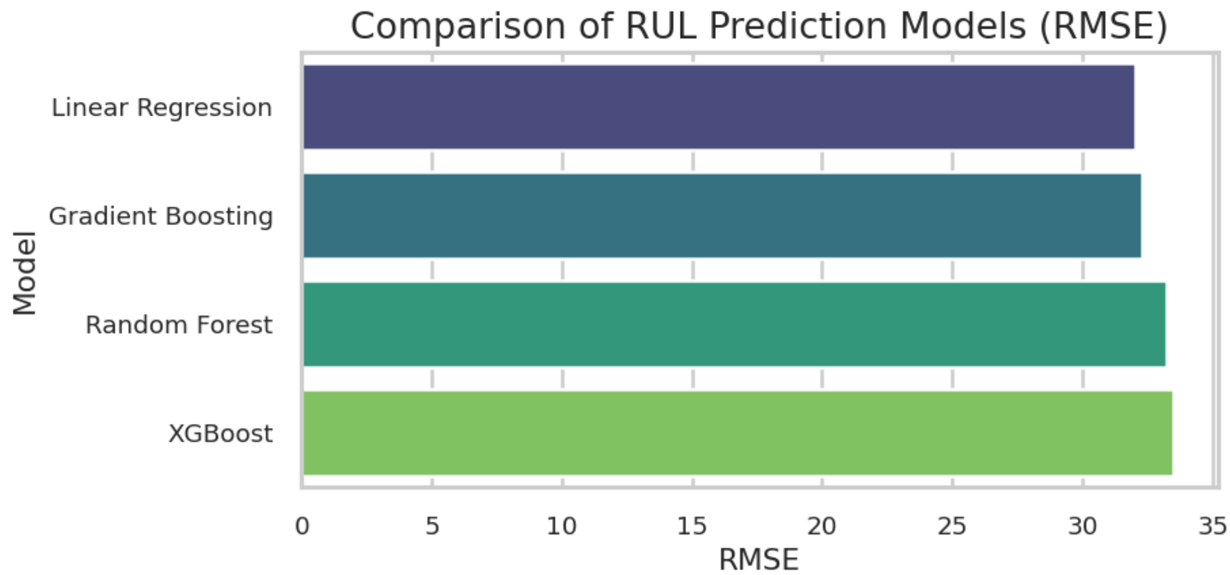

Figure 14: Comparision of RUL Prediction Models (RMSE)

4.2 Comparative Analysis of Models

The results indicate that all models performed within a narrow range, with RMSE values between **32.04 and 33.52**. This suggests that the FD001 dataset has relatively structured degradation patterns that can be captured even by simpler models.

Interestingly, Linear Regression outperformed more complex models in RMSE. This indicates that the relationship between sensor features and RUL is not highly nonlinear in this dataset. However, Gradient Boosting showed better MAE performance, meaning it handled prediction deviations more consistently.

4.3 Impact of Feature Engineering

Feature engineering was applied using lag features and rolling mean features to capture temporal patterns. The performance after feature engineering is shown in Table 2.

Table 2: Performance After Feature Engineering

Model	RMSE	MAE
Random Forest + FE	35.4834	25.1431
XGBoost + FE	34.1502	23.9159

The results show that feature engineering **did not improve performance**. In fact, RMSE increased for both models, indicating that the engineered features introduced noise or redundancy rather than useful information.

This suggests that the original dataset already contains sufficient information, and simple models can capture the underlying patterns effectively.

4.4 Hyperparameter Tuning Results

Hyperparameter tuning was applied to the XGBoost model using Randomized Search. The best parameters identified were:

- Number of estimators: 200
- Maximum depth: 4
- Learning rate: 0.03
- Subsample: 0.7
- Column sample by tree: 0.8

The performance of the tuned model is shown in Table 3.

Table 3: Tuned Model Performance

Model	RMSE	MAE
Tuned XGBoost + FE	32.4804	23.4990

The tuned XGBoost model achieved an RMSE of **32.48**, which is an improvement compared to the untuned XGBoost model (33.52). However, it still did not outperform Linear Regression in RMSE.

This shows that hyperparameter tuning provides **incremental improvement**, but does not drastically change model performance.

4.5 Final Model Comparison

The final comparison of models based on RMSE is as follows:

Table 4: Final Model Comparision

Rank	Model	RMSE
1	Linear Regression	32.04
2	Gradient Boosting	32.32
3	Tuned XGBoost + FE	32.48
4	Random Forest	33.21
5	XGBoost	33.52

The results clearly show that **Linear Regression is the best-performing model in this study**, followed closely by Gradient Boosting.

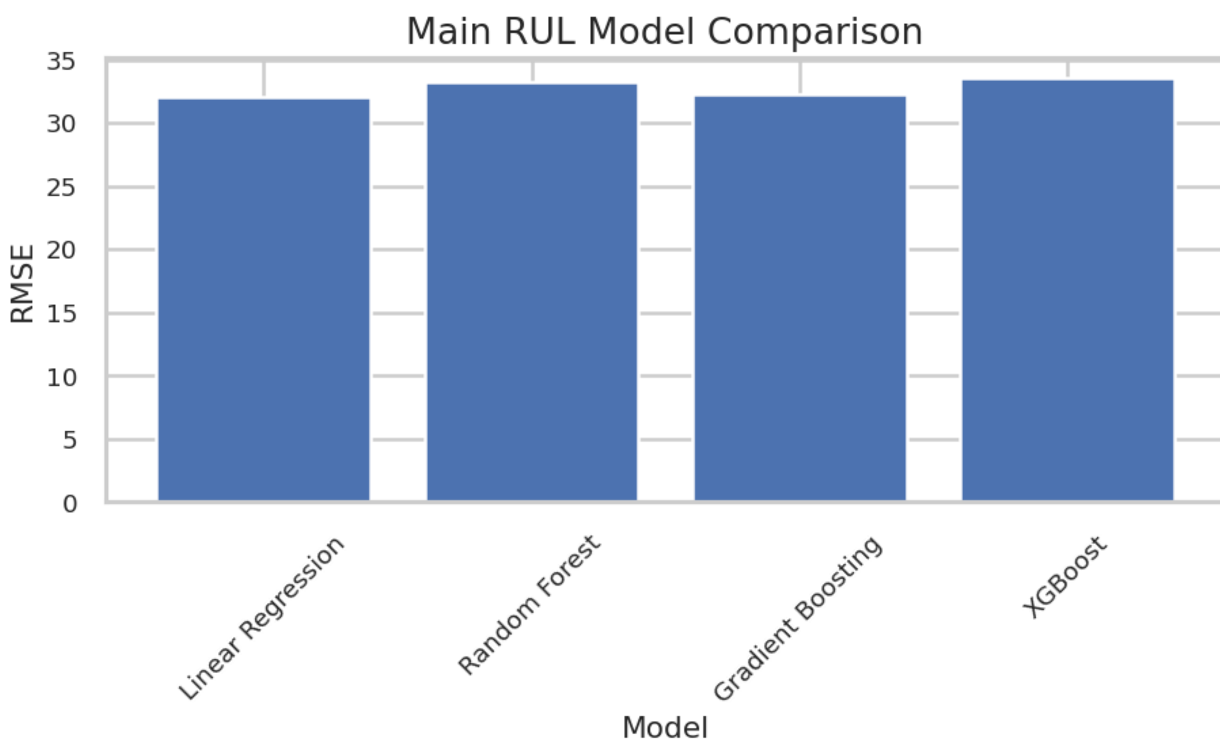

Figure 15: Main RUL Model Comparision

4.6 Anomaly Detection Results

Anomaly detection was performed using PCA and Isolation Forest.

The PCA-based method used reconstruction error as an anomaly score. It was observed that anomaly scores increased as engine cycles progressed, indicating that abnormal behaviour becomes more prominent near failure.

Similarly, Isolation Forest assigned anomaly scores that reflected deviations from normal behaviour. The model identified a small percentage of data points as anomalies, consistent with the defined contamination level of 5%.

Both methods showed that anomaly scores increase with engine degradation, confirming that sensor patterns change significantly as failure approaches. However, since the dataset does not include true anomaly labels, these results are exploratory and cannot be quantitatively validated.

5 Conclusion

This project focused on predicting the Remaining Useful Life (RUL) of aircraft engines and identifying anomaly patterns using multivariate sensor data from the NASA CMAPSS dataset. A complete data science pipeline was implemented, including data preprocessing, feature engineering, model development, and anomaly detection.

The results indicate that Linear Regression achieved the best performance in terms of RMSE, while Gradient Boosting achieved the lowest MAE. This suggests that the dataset exhibits relatively structured and partially linear relationships between sensor features and engine degradation. More complex models such as Random Forest and XGBoost did not significantly outperform simpler models, highlighting that increased model complexity does not necessarily lead to improved performance.

Feature engineering techniques, including lag features and rolling statistics, did not enhance model performance and, in some cases, increased prediction error. This implies that the original features already captured most of the relevant information, while additional transformations may have introduced redundancy or noise. Similarly, hyperparameter tuning of XGBoost resulted in only marginal improvements, reinforcing the observation that model performance is strongly influenced by the characteristics of the dataset.

Anomaly detection methods using PCA and Isolation Forest successfully identified abnormal patterns in engine behaviour. Both approaches demonstrated increasing anomaly scores as the engine approached failure, indicating that degradation patterns can be effectively captured using both statistical and machine learning techniques. However, due to the absence of ground truth anomaly labels, these results remain exploratory.

5.1 Limitations

Despite the promising findings, several limitations should be acknowledged. Firstly, the dataset used in this study is simulated and may not fully represent real-world engine behaviour. The FD001 subset contains only a single operating condition and fault mode, which limits the generalizability of the results to more complex real-world scenarios.

Additionally, the models implemented in this project are primarily static and do not fully capture temporal dependencies inherent in time-series data. Although feature engineering was used to incorporate some temporal information, it may not be sufficient to model long-term dependencies effectively. Furthermore, the lack of labelled anomaly data restricts the ability to quantitatively evaluate the performance of anomaly detection methods.

5.2 Future Work

Future work can address these limitations by utilising more complex datasets that include multiple operating conditions and fault types, thereby improving the generalizability of the models. Advanced deep learning techniques, such as Long Short-Term Memory (LSTM) networks or other sequence-based models, can be explored to better capture temporal dependencies in sensor data.

In addition, more sophisticated feature engineering methods and domain-specific knowledge can be incorporated to enhance model performance. For anomaly detection, future research can focus on using labelled datasets or developing evaluation frameworks for unsupervised methods to improve reliability and validation. Integrating predictive modelling with real-time monitoring systems could further enhance the practical applicability of predictive maintenance solutions.

‘

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7 Appendix

7.1. Final Project Code:

[Project Code]

7.2. Plots.

7.2.1

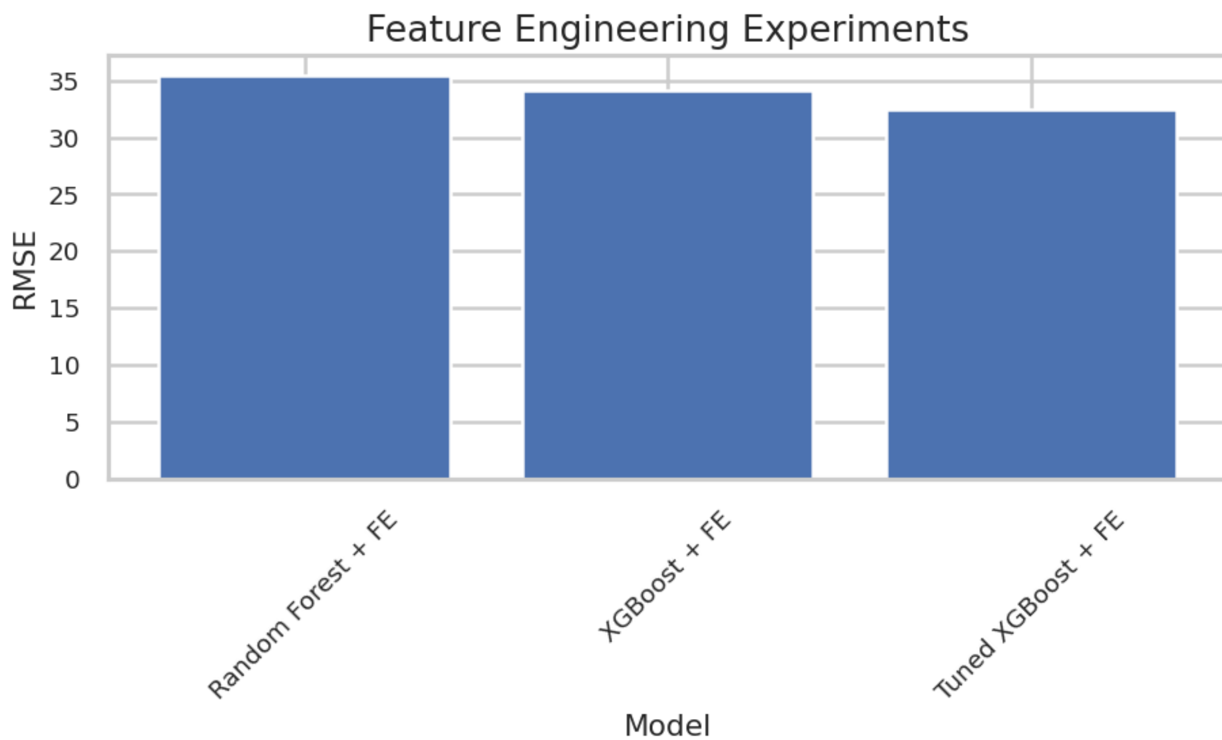


Figure: Feature engineering result comparison

7.2.2.

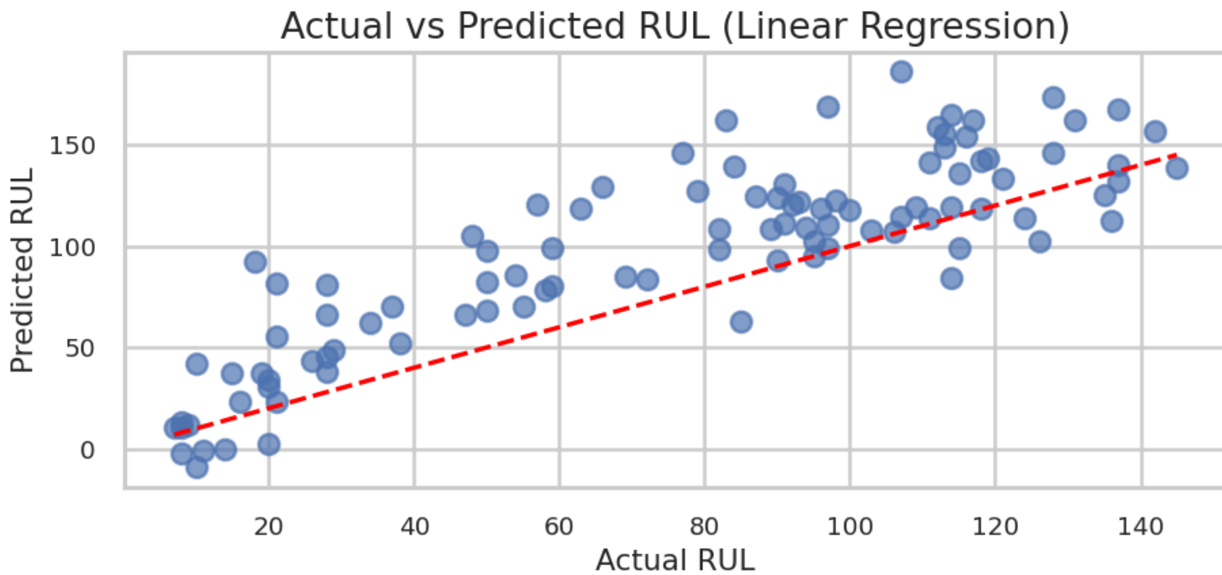
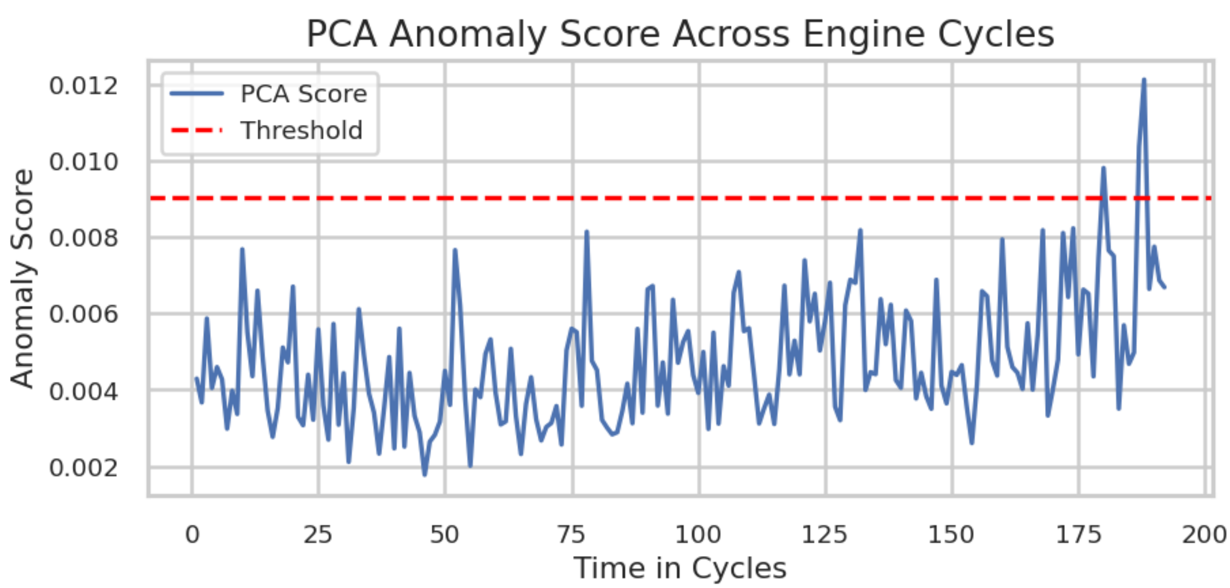


Figure: Actual vs Predicted RUL

7.2.3.



PCA Anomaly detection

7.3. Hyperparameter Tuning Results:

Fitting 3 folds for each of 10 candidates, totalling 30 fits

Best Params: {'subsample': 0.7, 'n_estimators': 200, 'max_depth': 4, 'learning_rate': 0.03, 'colsample_bytree': 0.8}

Tuned XGBoost RMSE: 32.48035952514708

Tuned XGBoost MAE: 23.498964309692383

7.4. Results Table.

	Model	**RMSE**	**MAE**
0	Linear Regression	32.042741	25.593806
1	Gradient Boosting	32.316593	23.221215
2	Random Forest	33.213210	24.446951
3	XGBoost	33.516057	24.027349